

ADS2001

STOCK PORTFOLIO OPTIMISATION PROJECT

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Executive Summary

Problem

The Standard and Poor's 500, more commonly known as the S&P 500, is a well-known stock market index that tracks the performance of 500 leading companies listed on stock exchanges in the United States. It is considered a reliable benchmark for overall market health and investment performance. The goal of this project was to create a collection of stocks, more commonly known as a stock portfolio, which could outperform the S&P 500, using only historical stock price data for model training. This posed the challenge of identifying the right combination of stocks from a large selection, without relying on future data. The project required developing a model that would select high-performing stocks while managing risk, using previous historical trends to achieve the highest possible returns.

The Data

For our analysis, we worked with a dataset sourced from Kaggle. The dataset provided us with the stock ticker symbol, the date, and the stock's adjusted closing price. This dataset gave us enough information to evaluate company-level and industry performance, detect trends, and engineer features such as percentage return, volatility, and correlation between stocks.

Findings

After testing several strategies, our key finding was that the optimal portfolio size was eight stocks. We evaluated three different portfolio weighting methods:

- A static approach, which yielded a return of 34%
- A dynamic non-machine learning strategy outperformed the others with a return of 37.78%
- A machine learning based approach, which significantly underperformed, returning -3.85%

These results suggest that the optimal approach is to use dynamic weighting, which automatically adjusts every three months with an eight-stock portfolio.

Introduction

What is the S&P 500?

The S&P 500 is a market-capitalisation-weighted index tracking 500 of the largest publicly traded companies in the United States. It is considered “one of the best gauges of large U.S stocks” (Kenton, 2024) due to the fact it covers “approximately 80% of available market capitalization” (S&P Global, 2025), and is used as a benchmark for stock market performance, including companies from a variety of sectors, including technology, financials, healthcare and more. Companies are selected for inclusion based on size, liquidity, and sector representation, meaning that they must be a large enough company, usually requiring a market capitalisation of several billion dollars, the company’s stock must be easy to trade in large volumes without affecting the price too much, and the index covers a wide range of industries to ensure that it reflects the overall U.S. economy, not just one sector. In doing so, the S&P 500 is well diversified, and as such, beating its performance consistently is a significant challenge for any investment strategy. This is demonstrated by the fact that in 2024, “65% of all active large-cap U.S. equity funds underperformed the S&P 500”, which was just worse than the “64% average annual rate” over the past 24 years. (Ganti et al., 2025).

Our Data

For this analysis, we used a dataset sourced from Kaggle, which contained historical stock data for companies listed on the S&P 500. The dataset included three variables:

- Ticker - the unique stock symbol for each company
- Date - the trading date in YYYY-MM-DD format
- Price - the adjusted closing price of the stock on that particular date

The dates given spanned across 9459 days, starting from the seventh of September 1993, up until the 31st of July 2019. The prices of the stocks only appeared if the stock was in the S&P 500 during that time, resulting in null values for the stock prices of several hundred stocks throughout the time period.

This dataset allowed us to track the company’s stock performance over time, calculate returns, and analyse trends. We were able to calculate the percentage change returns of the stocks over periods of time, the risk and volatility of stocks, and utilised this information to select a high-performing portfolio. We decided to focus on training the models using the data from 2010 to 2017, then predicting the future values, and comparing our portfolio to the actual performance of the S&P 500 during 2018 and 2019.

Data Quality and EDA

Data Cleaning and Exploratory Data Analysis

Exploratory Data Analysis (EDA) was a foundational step in this project, serving to not only familiarise ourselves with the dataset but also uncover key insights, identify trends and outliers, and prepare the data for advanced portfolio construction and optimisation.

The dataset provided consisted of closing adjusted prices for over 1,100 publicly listed companies, spanning from 1993 to 2019. For the purpose of this project, the training data was constrained to the period between 2010 and 2017, a range that reflects a full economic cycle and provides sufficient coverage across 2,922 trading days.

Data Cleaning & Data Quality:

The EDA process began with rigorous data cleaning and transformation. The raw CSV file was loaded, and the 'Date' field, which was originally displayed in YYYYMMDD integer format, was converted into a standard datetime object. This allowed for more intuitive time-based filtering, indexing and resampling operations, which are crucial when working with financial time series data.

To ensure analytical integrity and to avoid any biases, we opted to completely exclude companies that lacked data for even a single trading day. Only companies with full data coverage (i.e., non-null entries) for every trading day within the target period were retained. This reduced the original dataset to a more manageable and consistent set of stocks, each with a continuous price history, that enabled accurate calculations of returns and volatility.

Another important data quality consideration was both the format and structure of the data. The original column names included redundant suffixes, such as 'US Equity,' which were removed to improve readability and to ensure compatibility with the automated analysis tools.

Once the dataset was cleaned and validated, the next step involved generating descriptive statistics and computing total returns over the time frame. Using the first and last available stock prices, we calculated the percentage change in stock price for each company, thereby ranking them accordingly. This revealed a significant spread in performance across corporations, with several standout performers emerging. Companies such as Netflix (NFLX), BSET Furniture (BSET) and Regeneron Pharmaceuticals (REGN) demonstrated particularly strong growth, pointing toward high-performing sectors like communication services, healthcare and consumer cyclical.

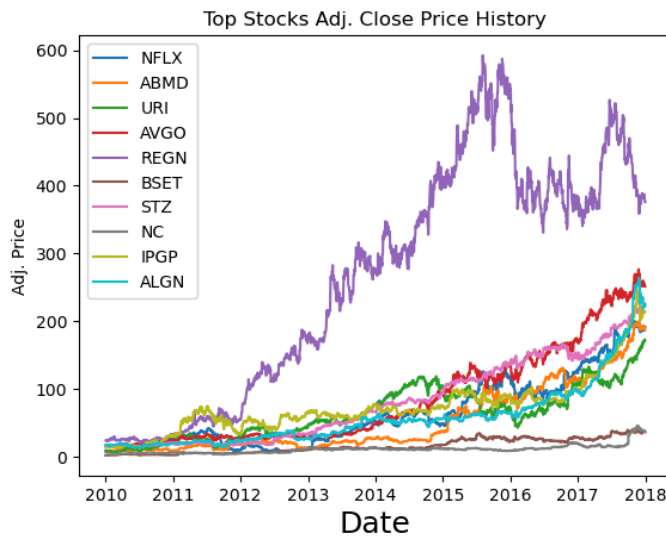


Figure 1. Top 10 performing stocks from 2010 to 2017

Outliers:

A particularly notable observation from the return distribution was the presence of a major outlier, **Regeneron Pharmaceuticals (REGN)**. As visualised in the performance graph of total returns across the selected companies, REGN demonstrated an exceptionally high percentage increase compared to its counterparts. Whilst most companies displayed moderate to strong growth over the 2010-2017 period, REGN's return was disproportionately high, standing well above the rest of the distribution. This exceptional growth can be largely attributed to the commercial success of the *Eylea Injection*, an eye drug, which was approved in 2011, for the treatment of patients with a neovascular form of age-related macular degeneration (Regeneron, 2012). The market response to this breakthrough was substantial, triggering a rapid rise in Regeneron's stock price and setting it apart from the broader distribution of returns. Identifying REGN as an outlier was critical, as it highlights how stock-specific events, such as successful drug development and FDA (Food and Drug Administration) approval, can dramatically influence stock performance. From a portfolio optimisation perspective, this emphasises the importance of balancing high-growth opportunities with diversification, as overexposure to such outliers can increase portfolio risk and also distort performance metrics.

In addition to individual stock analysis, a sector-level breakdown was also performed to evaluate the broader performance patterns across industries. Using *yfinance*, companies were grouped based on their sector classifications, and average returns were computed for each group over the target period. This revealed clear distinctions in sector performance. **Communication Services, Consumer Cyclical and Healthcare** sectors emerged as top performers, reflecting strong innovation cycles, increased consumer demand and also the commercial success of key corporations like Netflix and Regeneron.

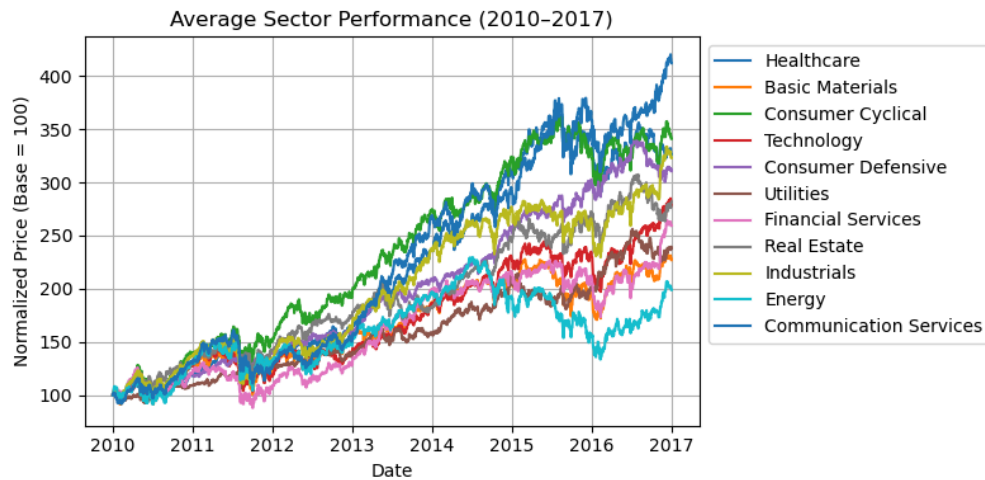


Figure 2. Average Sector Performance from 2010 -2017

By analysing sector-level trends, the EDA provided essential context for understanding relative risk and opportunity across sectors. It reinforced the importance of strategic diversification and underscores the value of sector-based filtering when building portfolios aimed at achieving superior risk-adjusted returns.

Model Development and Results

Overview

When constructing a portfolio, there are two main aspects to consider. The choice of stocks to hold and their respective weights. Several methods to determine the choice of stocks were examined. Given the difficulty in predicting the prices of stocks, a regression algorithm was not assessed. Instead, a classification approach was taken using gradient boosting to predict whether the stocks will net a high return or not. Another approach was through a simpler function defined to sort stocks based on their Sharpe Ratio over the training period.

Once the stocks had been chosen, the next problem was to determine how much to invest in each one. Two strategies were compared, namely static weights and dynamic weights. Static weights consist of deciding on the proportion of wealth invested in each stock and will remain constant throughout the whole testing period. On the other hand, a dynamically weighted portfolio will update the weight of each stock every 3 months based on its recent performance. In addition, to aid in simplicity, a constraint was added that restricted the use of negative weights (only long positions allowed).

There are several objective functions to consider when calculating the optimal weights for each stock. For example, the weights may be determined based on whichever combinations gave the lowest risk or, conversely, whichever had the highest expected return. This project focuses on maximising the Sharpe Ratio as the objective function. The Sharpe Ratio is a number that measures the risk-adjusted return of a given portfolio (CFI, 2025). It is calculated by dividing the expected return of the portfolio by its volatility (typically represented by standard deviation). Therefore, in order to maximise the Sharpe Ratio, the portfolio should aim to maximise expected return whilst minimising risk.

Gradient Boosting Classification Model

To investigate the best choice of stocks in a portfolio that will generate a high return for an investor, the first step involved preparing the training and testing data frames for the classification model. This was performed by taking 200 portfolios, where each portfolio consisted of 10 randomly selected stocks. Each portfolio was inputted into a function, which returned the yearly percentage Sharpe-weighted portfolio growth. The training data frame consisted of years from 2010-2017, and the testing data frame was from 2018-2019. These years were selected because they are recent and will help to evaluate the model's future predictive capability. Furthermore, they also simulate a realistic situation where investors make financial decisions based on past data. The structure of the training and testing data frames can be seen below.

Date	1	2	3	4	5	6	7	8	9	...	191	192	193	194	195
2010	0.695281	0.461721	0.318971	0.950866	0.589436	0.665475	0.403906	0.796153	0.550049	...	0.432096	0.512312	0.598258	0.484074	0.477618
2011	0.985380	0.325819	0.610970	0.608272	0.433759	0.451024	-2.619223	-0.537587	0.651059	...	0.426159	0.723133	0.769889	-1.169309	-1.747712
2012	0.310837	0.311417	0.236587	0.245464	0.406309	0.474165	0.544867	0.541714	0.251405	...	0.449839	0.743014	0.598202	0.488854	0.488927
2013	0.392819	0.359852	0.421340	1.045875	1.059273	0.387303	0.589638	0.521497	0.521572	...	0.354379	1.582345	0.480093	0.515852	0.815014
2014	0.413535	0.884996	0.333466	0.547347	0.398711	0.317306	1.021508	0.452994	0.196338	...	0.323469	0.394202	0.634688	0.322801	0.634653
2015	-0.386523	-0.858562	0.379845	1.385799	-4.878755	-0.523913	-0.735612	-0.512158	-1.391575	...	0.597756	-0.565397	-0.945585	0.361377	0.400228
2016	0.602418	0.502708	0.178115	0.399876	0.479205	0.710861	0.621330	0.583976	0.387294	...	0.311650	6.382776	0.346820	-1.524291	-2.859784
2017	1.605691	0.509812	0.323056	0.461234	0.988155	1.967553	-94.499937	0.528235	0.236745	...	0.396477	1.371358	0.674273	0.417513	0.505616

rows × 201 columns

Table 1. Training Data Frame

Date	1	2	3	4	5	6	7	8	9	...	191	192	193	194
2018	-0.388507	-0.451245	-0.318854	-0.340464	1.894951	-0.348403	-10.839397	-0.366594	-0.533339	...	-0.613947	-0.246999	-0.318199	-1.105018
2019	0.448536	0.310702	-0.001245	-0.003685	0.243959	0.750339	-0.002702	0.316810	-0.002180	...	0.289444	0.341937	0.559462	-1.640276

rows × 201 columns

Table 2. Testing Data Frame

However, the model also requires information on stocks in each combination to learn about the favourable ones. To provide this, both data frames were manipulated so that all feature variables are individual stocks instead of unique portfolios. The target variable is 'Return', where portfolios are classified as 'high return' if the percentage Sharpe-weighted portfolio growth is more than 10%, and 'not high return' if this requirement is not met. Below is the result of making these changes to the training and testing data frames.

	CI	AKAM	INCY	GRMN	FOSL	CTAS	CCK	VTR	BSX	PNW	...	AMG	MXIM	UPS	JCP	XOM	MHK	AIZ	NYT	APD	SSP
0	1.0	1.0	1.0	1.0	1.0	1.0	1.0	1.0	1.0	1.0	...	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
1	1.0	1.0	1.0	1.0	1.0	1.0	1.0	1.0	1.0	1.0	...	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
2	1.0	1.0	1.0	1.0	1.0	1.0	1.0	1.0	1.0	1.0	...	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
3	1.0	1.0	1.0	1.0	1.0	1.0	1.0	1.0	1.0	1.0	...	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
4	1.0	1.0	1.0	1.0	1.0	1.0	1.0	1.0	1.0	1.0	...	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...
1595	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	...	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
1596	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	...	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
1597	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	...	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
1598	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	...	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
1599	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	...	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0

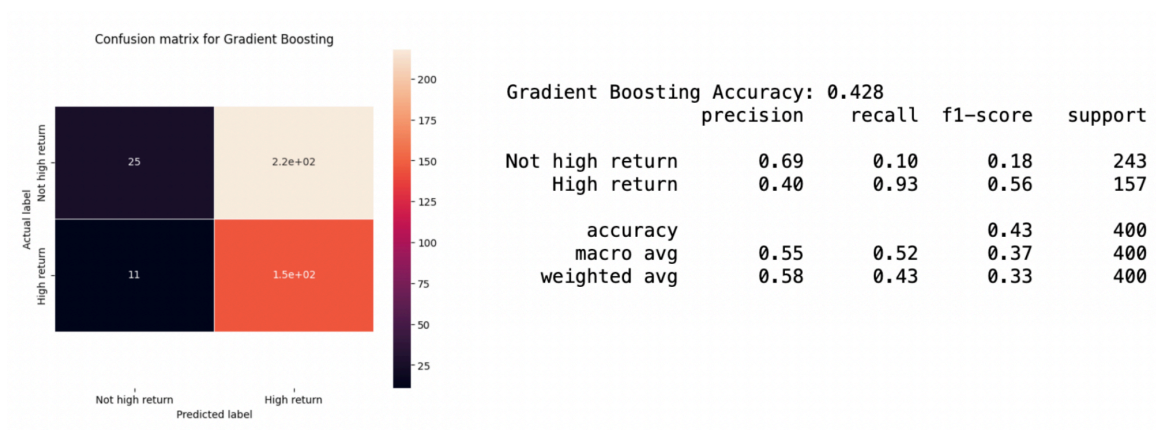
Table 3.Final Training Data Frame

	CCL	AIZ	HIG	FTV	FCX	COG	CAL	TRIP	BLK	WDC	...	GM	MAA	EQIX	BHGE	AMAT	MDLZ	TDC	IBM	VAR	LNC
0	1.0	1.0	1.0	1.0	1.0	1.0	1.0	1.0	1.0	1.0	...	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
1	1.0	1.0	1.0	1.0	1.0	1.0	1.0	1.0	1.0	1.0	...	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
2	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	...	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
3	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	...	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
4	0.0	0.0	0.0	0.0	0.0	0.0	0.0	1.0	0.0	0.0	...	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...
395	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	...	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
396	0.0	0.0	0.0	1.0	0.0	0.0	0.0	1.0	0.0	0.0	...	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
397	0.0	0.0	0.0	1.0	0.0	0.0	0.0	1.0	0.0	0.0	...	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
398	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	...	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
399	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	...	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0

400 rows × 613 columns

Table 4. Final Testing Data Frame

A Gradient Boosting classification model was used to predict the choices of stocks that deliver high returns, with the confusion matrix and classification report shown below.



Result 1. Confusion Matrix and Classification Report of Gradient Boosting Model

The model's strength lies in correctly identifying a large number of actual high-return portfolios, which is evident by the high recall score of 0.93. However, the model also has a low precision score for high return (0.40), which suggests that many predicted high-return portfolios did not actually give a high return (hence many false positives). This may be due to possible overfitting on the training data, meaning that an assumption made by the model is that the relationships seen in 2010-2017 would also hold in 2018-2019.

On the basis of this limitation and the assumption made by the Gradient Boosting model, it is recommended to use the model for shortlisting potential high-return portfolios and not to decide on the best stocks to invest in. An attempt was made to decide the best stock portfolio for investment based on this model, by using a non-traditional optimisation strategy. This involved examining the probabilities that the model classifies high returning portfolios, and determining the portfolio that was classified in this class with the most confidence. The recommended portfolio and the probability of its predicted classification are displayed below.

```
probs = []
for i in range(0,400):
    sample_input = X_test.iloc[[i]]
    probs.append(predict_high_return_probability_gradient_test(sample_input))
ind = probs.index(max(probs))
stock_lst = []
c = 0
for s in X_test.iloc[ind]:
    if s == 1:
        stock_lst.append(X_test.iloc[ind].index.tolist()[c])
    c += 1
print(stock_lst)
print(max(probs))
print(y_test[ind])
```

```
['MET', 'MCK', 'CAH', 'MOS', 'FLIR', 'M', 'ETN', 'PNC', 'DDS', 'HOG']
0.9464807650130063
```

Result 2. Determining the stock portfolio that was most confidently predicted as a high return

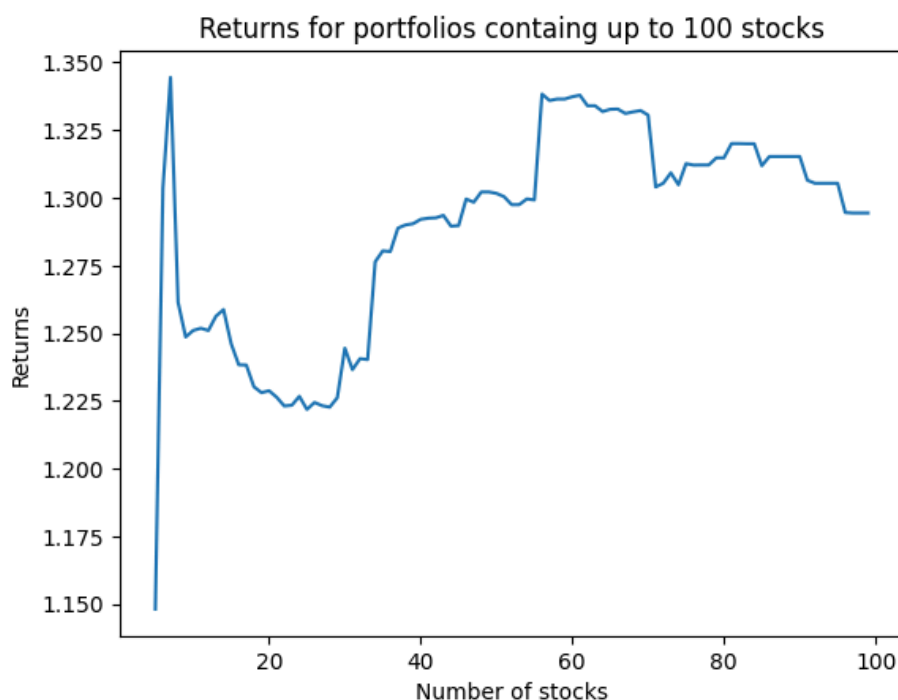
The actual return classification of the recommended portfolio was compared to the prediction, and due to high false positives, it was found that the portfolio was incorrectly predicted. Hence, we can conclude that using the model for determining the best portfolio can be misleading for investors.

Static Weights

In a portfolio consisting of static weights, the proportion of wealth invested in each stock is determined from the beginning, and the portfolio will remain untouched throughout the whole testing period. Since the objective function of the project was to maximise the Sharpe Ratio, the weights of each stock will be determined based on this.

To achieve this, the mean-variance optimiser (MVO) from the PyPortfolioOpt Python library was utilised. MVO is based on the modern portfolio theory, a mathematical framework which constructs portfolios that “maximises the expected return [for a] given level of risk” (Scott, 2023). The MVO functions by taking in as inputs: the expected returns of assets extrapolated from their historical prices, as well as the covariance matrix of assets’ past returns (measures how the assets perform in relation to one another) (Martin, 2018). It then solves a mathematical optimisation problem to find the best combination of weights that results in the highest expected return for a given level of risk. By passing the function ‘`ef.max_sharpe()`’, it returns an array of weights associated with the portfolio with the highest Sharpe Ratio.

Due to the poor performance of the stocks from the gradient boosting classification model, they were not considered. Instead, a function was defined that sorted all the stocks based on the Sharpe Ratios measured from their historical prices. To determine how many stocks should be held in the portfolio, a ‘for loop’ was used to analyse the performance of portfolios containing 5 stocks up to 100. During each iteration, the MVO was deployed to find the highest Sharpe Ratio portfolio for each basket of stocks.



As seen in Figure 3, having 8 stocks was deemed to be most optimal, these stocks being: 'NOC', 'MO', 'HD', 'LMT', 'EXR', 'TDG', 'CTAS' and 'STZ'. Thus, these were the stocks that made up the resulting portfolio, and their performance was further analysed.

Figure 3: A line graph representing the returns for a given number of stocks

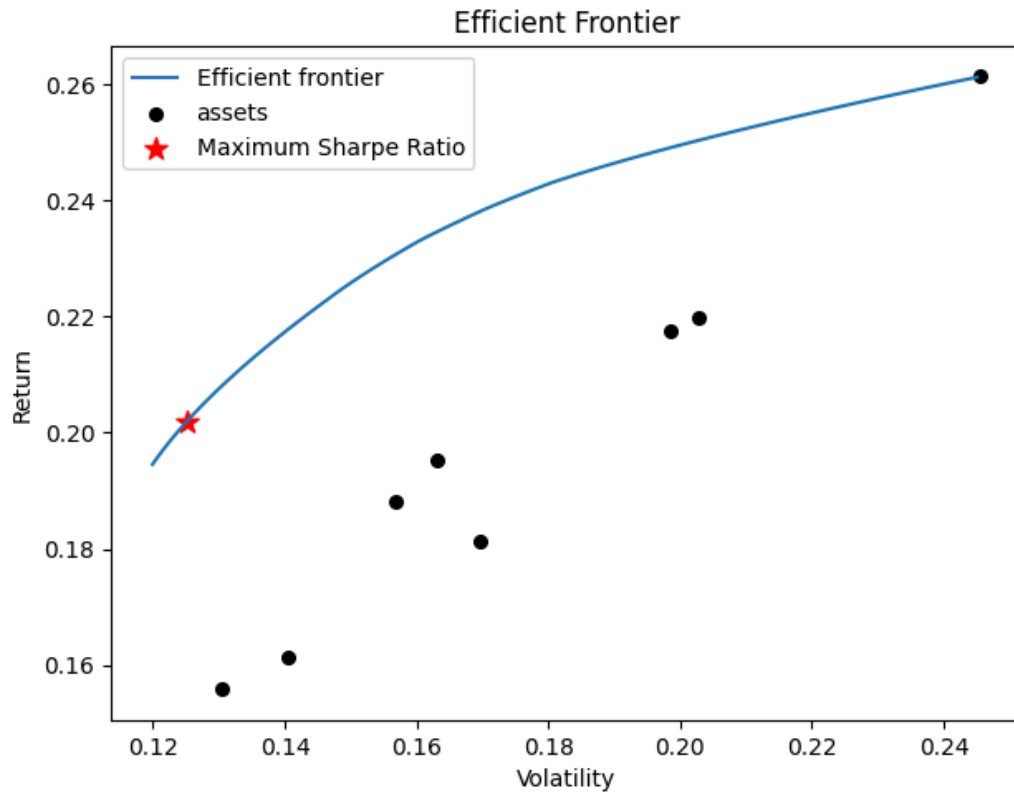


Figure 4: Efficient Frontier of portfolio

The MVO is also capable of plotting an efficient frontier, a line that represents the optimal portfolios for any given level of risk (Horti, 2023). The red star represents the portfolio that contained the highest Sharpe Ratio of 1.61 and was used for the project.

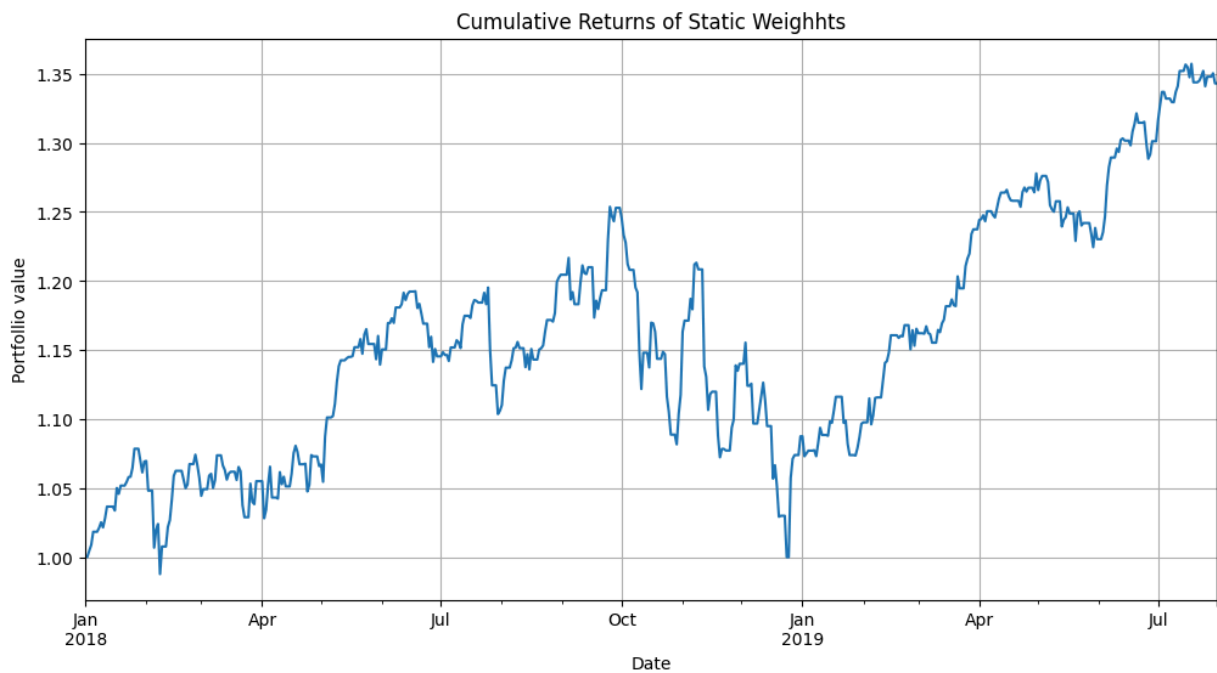


Figure 5: A line graph representing the cumulative returns for the optimal portfolio

The performance of the portfolio over the testing period (2018-mid 2019) can be seen in Figure 5. Despite some volatility and a few drops, it finished on an upward trend and ended with a 34.45% return.

When retrieving the portfolio with the highest Sharpe Ratio from the MVO, several of the weights in the portfolio were close to 0, some even at 0. As a result, some of the stocks considerably dominated and weighed upwards of 30–40%. Although this still resulted in the highest Sharpe Ratio, investors may still feel uneasy about having the majority of their portfolio composed of 1-2 stocks. As such, L2 regularisation was utilised to improve the MVO. It discouraged stocks from having very large weights by adding a penalty term and thus improved diversification and avoided unintuitive allocations. Doing so dropped the Sharpe Ratio of the portfolio from 1.61 to 1.56. However, despite the slightly lower Sharpe Ratio, the portfolio performed much better. A return of 34.45% was achieved (as seen in Figure 5) using L2 regularisation compared to a return of 23.49% without. The improvement in return despite the poorer Sharpe Ratio can be attributed to the decreased volatility in distributing investments across a greater quantity of stocks and therefore sectors. As referenced in our EDA, this ensures that the portfolio remains stable after considering poorly performing sectors.

Thus, the static weights strategy found the optimal number of stocks to be 8, which had a return of 34.45%. The corresponding weights can be found below in Figure 6.

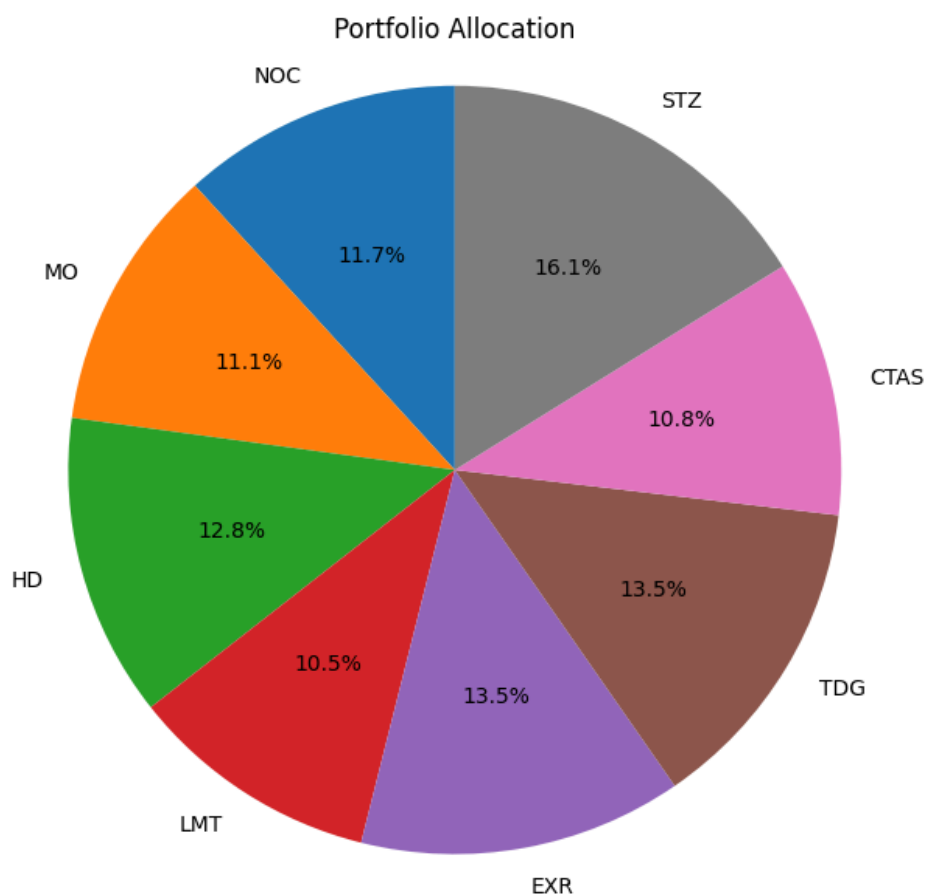


Figure 6: A pie chart representing the optimal portfolio and its respective weights

Dynamic Weights using Machine Learning

There were two strategies taken to analyse a dynamically weighted portfolio, with the first utilising machine learning. Rather than using machine learning to predict the prices, a more unconventional approach was followed, which used it to predict the optimal weights for a given basket of stocks. Doing so allowed for an easier comparison between the static and dynamically weighted strategies. The number and choice of stocks were kept constant, with the only differences being the method of determining the optimal weights as well as the frequency with which the weights were rebalanced.

A study by the J.P. Morgan AI Research team found that deep learning models were often used to predict the optimal weights (Sood et al., 2023). Thus, a neural network was used to achieve this. Neural networks are a type of machine learning algorithm that consists of three layers: input, hidden and output layers. The data is first received in the input layer, which is then processed in the hidden layer based on a custom function, with its result being returned in the output layer. This process of transporting the data from layer to layer is known as feeding forward (IBM, 2021). The network then evaluates its performance by comparing its outputs to the actual target values based on a custom loss function. In the context of this project, the input layer of the model took in the past year of returns and standard deviations for each stock. Since the objective function of the project was to maximise the Sharpe Ratio, the loss function was defined to minimise the negative Sharpe Ratio. The model will then output a set of weights for each stock.

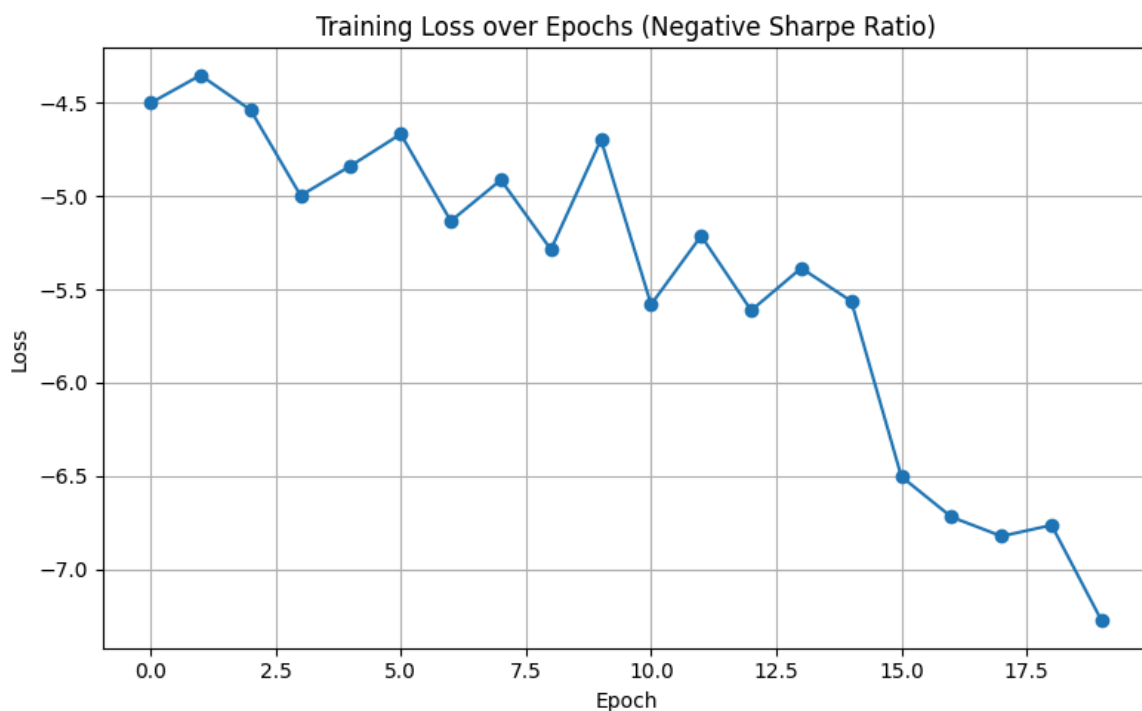


Figure 7: A line graph representing the cumulative training loss after each training iteration

As observed in Figure 7, the training loss is gradually decreasing during each iteration of training the model. Since the loss function is defined to minimise the negative Sharpe Ratio, minimising the loss results in maximising the Sharpe Ratio. Thus, having lower losses results in better model performance. However, the network did not perform well over the testing period.

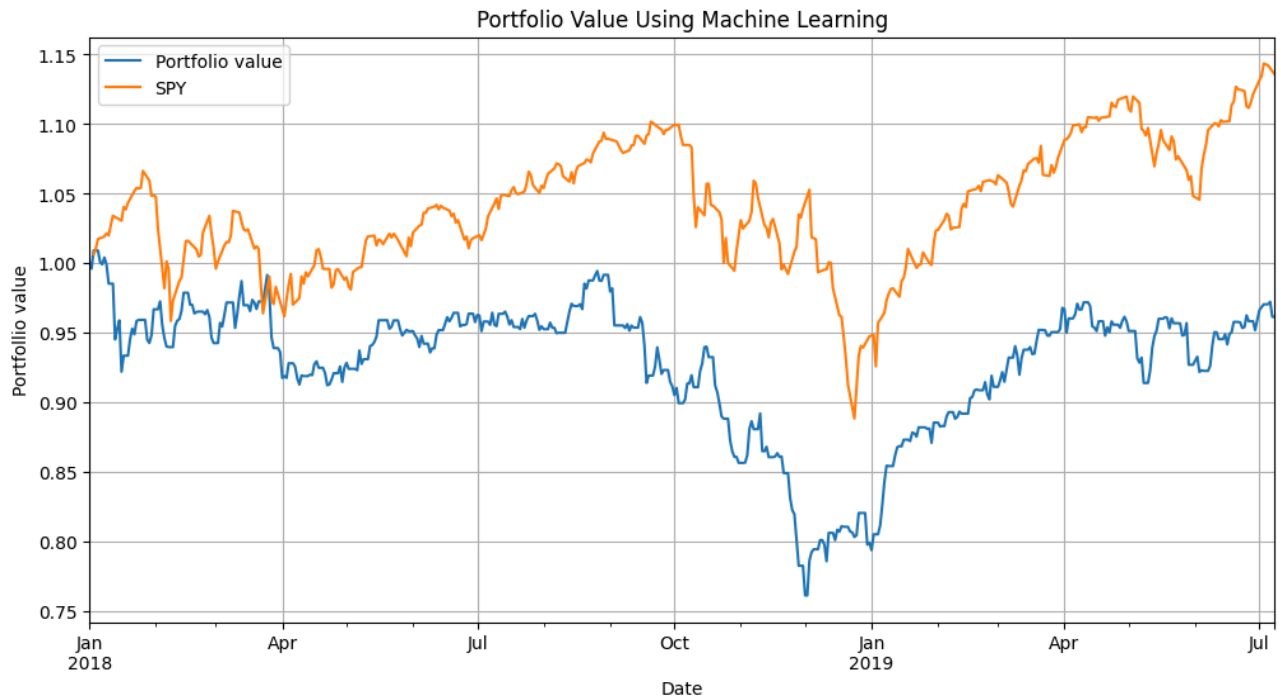


Figure 8: A time series graph representing the cumulative return of the dynamic weights portfolio compared to the SPY index

Figure 8 visualises the cumulative return of the dynamic portfolio over the testing period (blue line) alongside the SPY, an exchange-traded fund that tracks the performance of the S&P 500, which is also a commonly used benchmark (Nickolas, 2023). As seen, the portfolio performed much poorer compared to the SPY. In fact, it had a -3.85% return and had a -0.20 Sharpe Ratio.

Thus, the machine learning approach did not seem appropriate to conduct a dynamically weighted portfolio. However, several changes can be made to potentially improve the model. Firstly, the dimension of the hidden layer can be increased to increase the network's capacity to learn. This can be used in conjunction with higher training rounds. In addition, rather than relying on expected returns and standard deviations as the input, further feature engineering could have been implemented to add more financial indicators such as relative strength index, volume and moving averages.

Dynamic Weights without Machine Learning

The second strategy consisted of a much simpler optimisation method. Again, the same number of stocks (8) and choice of stocks were used for consistency. The frequency of portfolio rebalancing was also kept constant as before (3 months). Rather than using machine learning, the same mean-variance optimiser utilised in the static weights portfolio was applied. In essence, this strategy was almost identical to the static weights, with the only difference being that the weights were regularly updated.

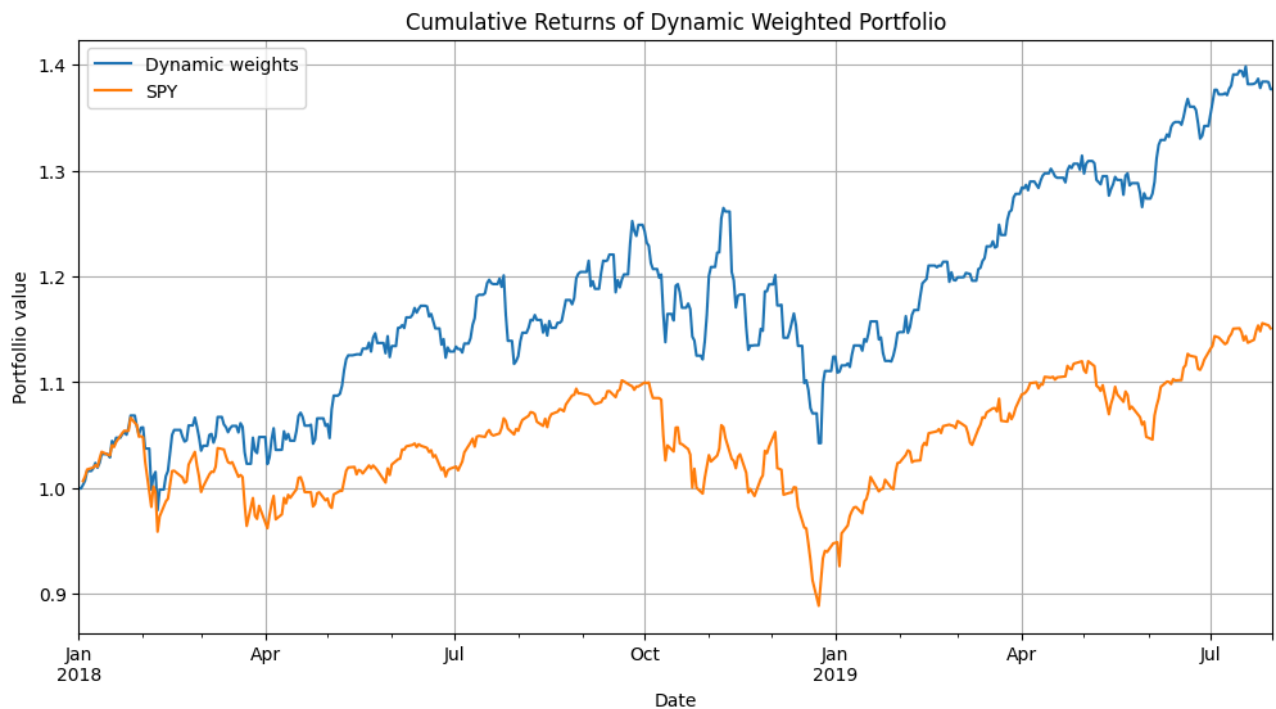


Figure 9: A time series graph representing the cumulative return of the dynamic weights portfolio (non-ML) compared to the SPY index

This time, as seen in Figure 9, the dynamic portfolio performed exceptionally well and even outperformed the SPY. Specifically, it netted a 37.78% return and had a Sharpe Ratio of 1.58. This portfolio had the highest return among all others.

Conclusion

This project set out to build a stock portfolio capable of outperforming the S&P 500 using only the historical stock price data provided. Through testing several portfolio strategies, we found that a portfolio of eight selected stocks delivered the highest returns.

After evaluating three different portfolio weighting methods, the dynamic non-machine learning based approach achieved the highest return of 37.78%, slightly outperforming the static weighting model at 34%. In contrast, the machine learning based approach had a return of -3.85%.

These results suggest that the optimal approach is to use dynamic weighting, which automatically adjusts every three months with an eight-stock portfolio. Whilst it outperformed the static weighting by 3.78%, this was before factoring in brokerage fees. Once factoring in the fees of trading stocks, the dynamic model only outperformed the static model by 1.78%.

However, there were several limitations to our project. Firstly, our analysis solely relied on the adjusted closing prices of the stocks, which limited the scope of our analysis. We did not include other important financial indicators such as the Relative Strength Index (RSI) or trading volume, which could have provided a more comprehensive view of stock performance and risk. Secondly, our model only used a testing period from 2018 to mid-2019, as well as was trained over eight years. This meant that we didn't have a comprehensive amount of time to determine whether the portfolio would perform well in the long term. Additionally, we focused on maximising the Sharpe ratio. While useful for measuring risk-adjusted returns, future work could explore other key performance metrics.

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